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CAMPUS CARE COMPLAINT MANAGEMENT SYSTEM

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MASTER OF COMPUTER APPLICATION

(COMPUTER TECHNOLOGY AND APPLICATIONS)

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ABSTRACT

The **Campus Care Complaint Management System** is an online web application developed to simplify and speed up how complaints are handled across a college campus. In colleges and hostels, students and staff often face common problems like non-working classroom projectors, broken computers in labs, Wi-Fi connectivity issues, electrical faults, cleanliness problems, or hostel water issues.

Traditionally, students had to register complaints in a paper register or report them verbally to staff and wardens. This traditional method causes many problems: complaints are often misplaced or forgotten, students have no way to know whether their problem is being worked on, and college authorities cannot easily see which problems are urgent.

To solve these issues, **Campus Care** provides a simple and transparent digital portal. Any student can log in using their college email, fill out a simple complaint form, select the issue category (such as Hostel, Wi-Fi, or Electrical), set an urgency level (Low, Medium, High), and submit it. The system automatically creates a unique tracking ticket number (for example, *CC-2026-0014*) and saves it safely in the database.

College administrators have a dedicated Admin Dashboard where they can see all complaints in one place, assign technicians to fix them, update the status to **In Progress**, and finally mark the complaint as **Resolved** with a note explaining the fix. Students can check their dashboard at any time to see the live status and complete history of their complaint. This system ensures total accountability, eliminates paperwork, and saves time for everyone on campus.

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Chapter 1

INTRODUCTION

1.1 Project Overview

In every college campus, maintaining classrooms, computer labs, libraries, hostels, electrical fixtures, and internet infrastructure is an ongoing everyday job. Whenever something breaks down—such as a fan stopping, a projector failing, or internet dropping—students need an easy way to report it to the college administration.

In most colleges, this is still done by writing in paper registers or telling staff verbally. These older methods are slow, complaints often get lost, and students never know when their issue will be fixed.

To solve this everyday problem, we built **Campus Care Complaint Management System**. It is a web application where students can submit complaints online 24/7, track the progress in real time, and administration can manage and resolve all campus maintenance issues from one single dashboard.

1.2 Key Objectives

The main goals of the Campus Care project are:

- **Online Complaint Registration:** Allow students to submit grievances anytime from their phone or laptop without visiting administrative offices.
- **Unique Ticket Tracking:** Automatically generate a unique ticket ID (like *CC-2026-0014*) for every complaint so it can never be lost.
- **Categorized Issues:** Group issues into 9 simple categories (Classroom, Lab, Wi-Fi, Electrical, Cleanliness, Library, Hostel, Administration, Other).
- **Priority Levels:** Let students set priority (Low, Medium, High) so emergency problems (like power cuts or water leaks) get handled first.
- **Status Transparency:** Show clear status badges: *Submitted* → *In Progress* → *Resolved* so students know exactly what is happening.
- **Admin Management Dashboard:** Give administrators a clear view of total complaints, active issues, and resolved tasks with simple filters.

1.3 Problem Statement

Traditional manual complaint methods in educational institutions have serious disadvantages:

1. **Lost and Forgotten Complaints:** Complaints written on paper slips or registers often get misplaced or ignored with no proof of submission.
2. **No Status Updates:** Students have no idea whether someone is working on their issue or who is assigned to fix it.
3. **Wasted Time:** Students and staff spend unnecessary time making repeated personal visits to offices to check complaint status.
4. **No Priority Handling:** Important safety hazards (like short circuits or open wires) sit behind routine requests because there is no automated priority filtering.
5. **Lack of Records:** College heads have no simple way to see monthly maintenance statistics, staff response times, or recurring repair issues.

1.4 Proposed Solution

Campus Care provides a complete digital solution that handles complaints from start to finish:

- **Step 1: Student Login:** The student logs in securely with their email and password. New students can register in 30 seconds.
- **Step 2: Submitting Complaint:** The student enters a title, chooses the problem category, sets priority (Low/Medium/High), types the location (e.g., *Lab 3*), and writes a brief description.
- **Step 3: Instant Ticket Creation:** The system instantly saves the complaint in the database, marks its status as **Submitted**, and generates a ticket number.
- **Step 4: Admin Review & Action:** The college administrator checks the new complaint on the Admin Dashboard and changes its status to **In Progress** while maintenance staff works on it.
- **Step 5: Resolution & Notes:** Once fixed, the admin marks it **Resolved** and adds remarks explaining what was done. The student sees the green checkmark and resolution note instantly on their screen.

Chapter 2

SYSTEM DESIGN

2.1 System Architecture

The **Campus Care** system is built using a modern 3-tier architecture that clearly separates the user interface, backend application logic, and database storage:

1. Frontend Layer (What Users See)

Built using **ReactJS** and styled with **Tailwind CSS**. This layer runs directly in the user's web browser. It is fully responsive, meaning it looks and works great on mobile phones, tablets, and desktop computers.

- **Login and Register Page:** Simple forms for logging in or signing up.
- **Student Dashboard:** Shows total complaints, active issues, and resolved tickets.
- **Submit Complaint Form:** Form with category pickers, priority options, and text fields.
- **My Complaints Page:** Shows a list of all complaints submitted by the student with colored status tags.
- **Complaint Timeline:** Shows the step-by-step history of status changes and admin notes.
- **Admin Dashboard:** Comprehensive overview with cards showing counts and filterable tables.

2. Backend Layer (Server & API Logic)

Built using **Node.js** and the **Express.js** framework. It receives requests from the frontend, checks security tokens, runs business rules, and saves or fetches data from the database.

3. Database Layer (Safe Storage)

Uses **MongoDB** with **Mongoose ODM**. It stores user profiles and all complaint details with an unalterable history log so that past records cannot be accidentally deleted or changed.

2.2 Use Case Diagram

The Use Case Diagram shows how different users (Student and Administrator) interact with the Campus Care system. Students can lodge complaints and track their status, while administrators can review complaints, update status, and view analytics.

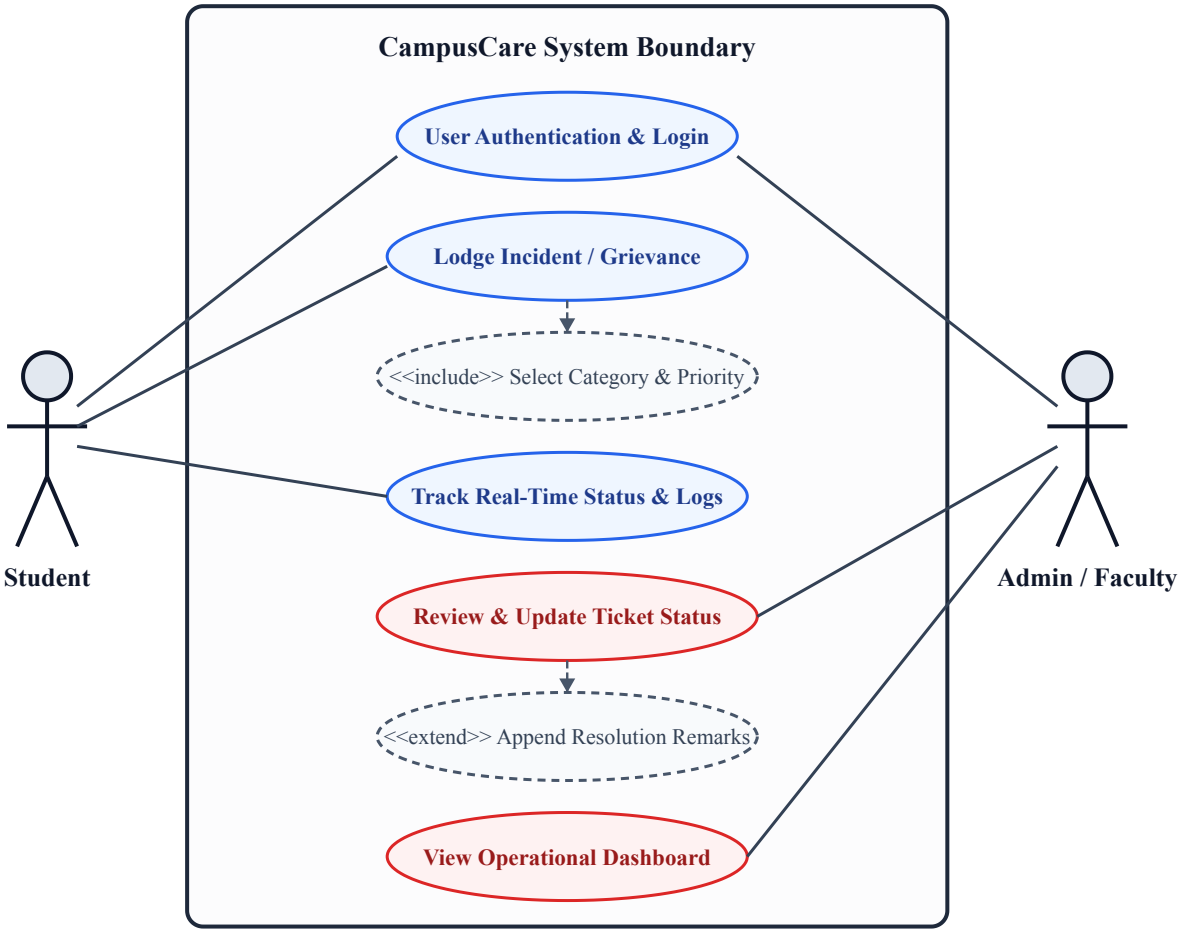


Figure 2.1: Use Case Diagram of Campus Care System

2.3 Data Flow Diagram (DFD)

The Data Flow Diagram illustrates how information flows through the system: starting from student authentication and complaint submission, storing records in the database, and updating the admin and student dashboards.

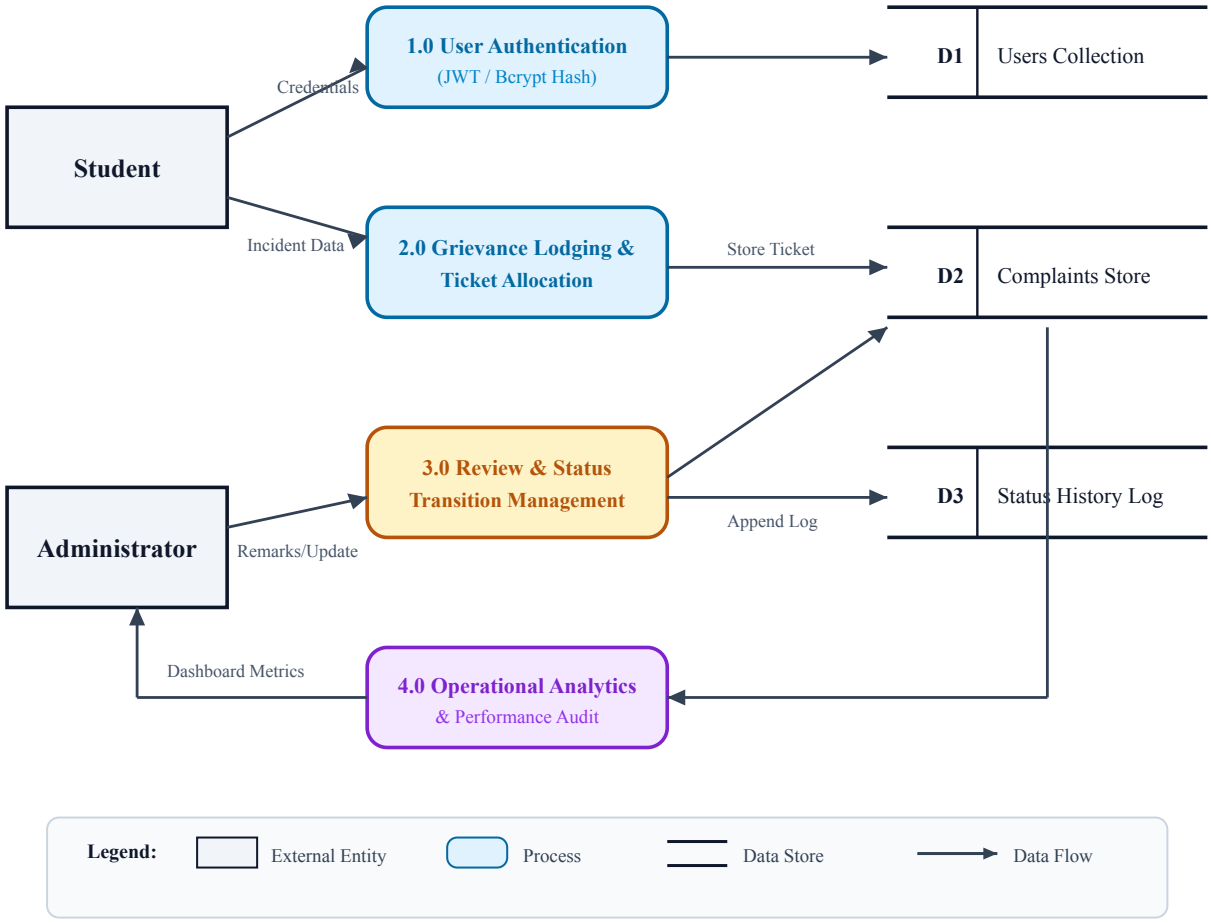


Figure 2.2: Data Flow Diagram (DFD) of Campus Care System

2.4 Class Diagram

The Class Diagram displays the main object structures in the codebase: the User class (Student/Admin roles), the Complaint class, embedded StatusHistory, and the Controllers that handle logic.

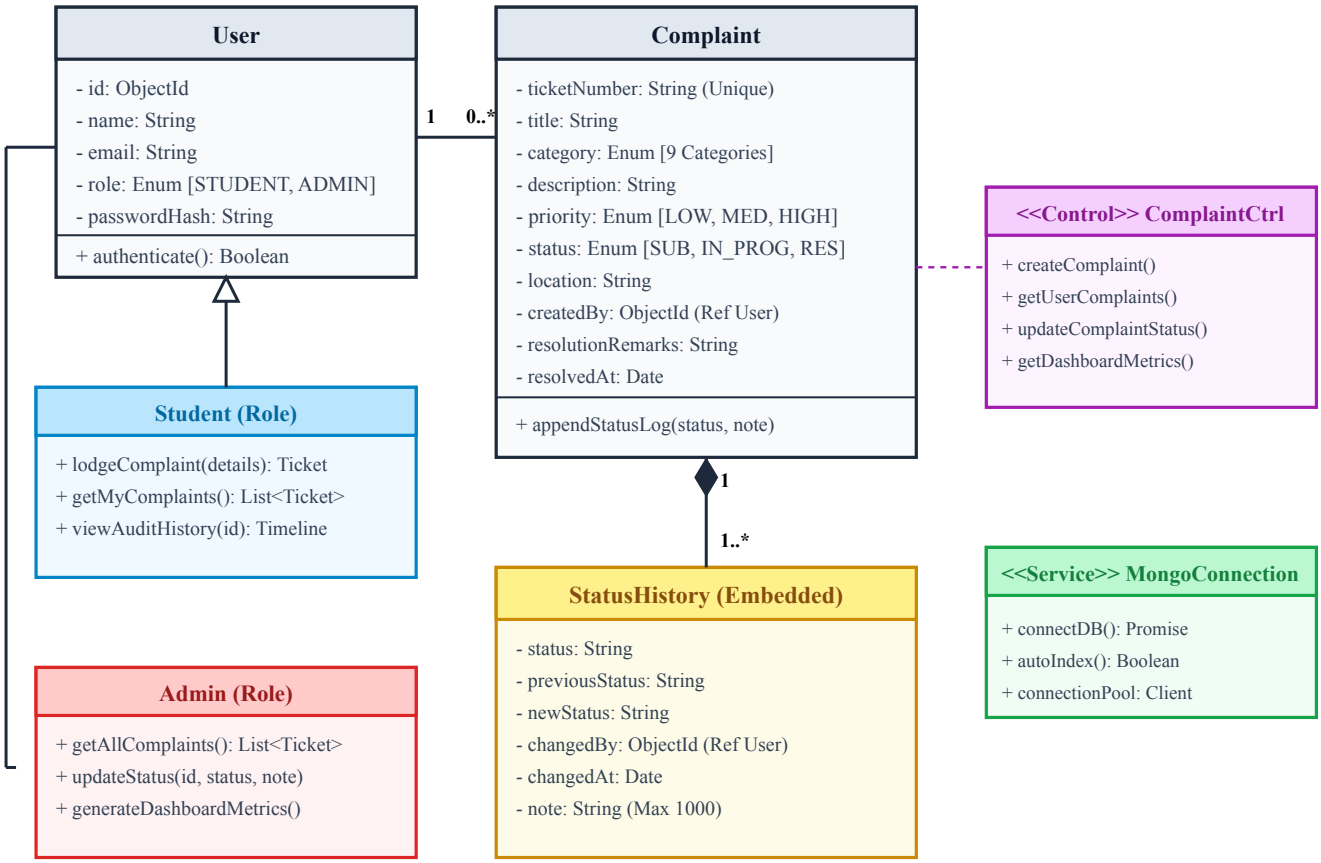


Figure 2.3: Class Diagram of Campus Care System

2.5 Activity Diagram

The Activity Diagram shows the step-by-step workflow: a student fills the form, the system validates inputs and creates a ticket, the admin reviews the complaint, updates status to In Progress, and marks it Resolved with notes.

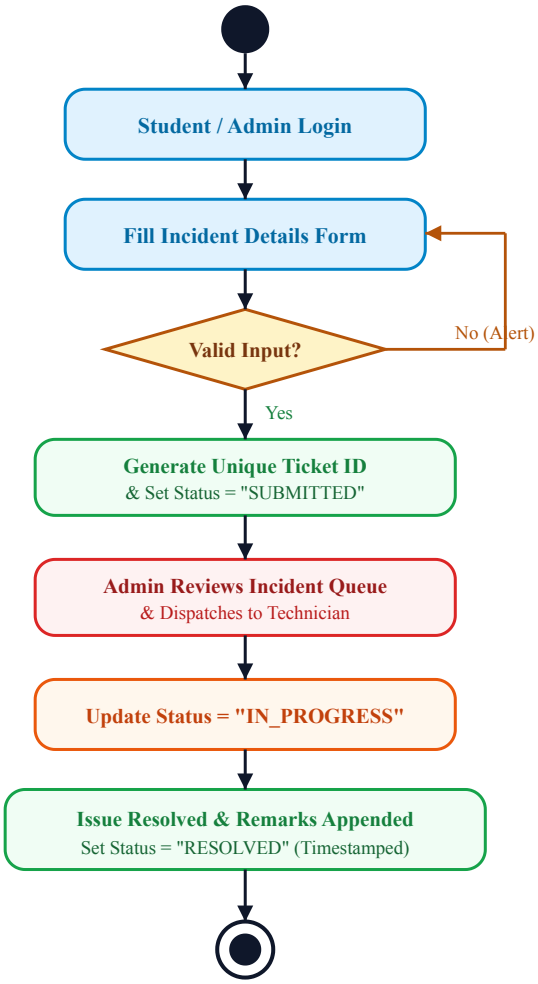


Figure 2.4: Activity Diagram of Complaint Lifecycle

2.6 Sequence Diagram

The Sequence Diagram illustrates the exact time-ordered messages sent between the Student, React Frontend, Express Server, MongoDB Database, and the Admin during complaint submission and resolution.

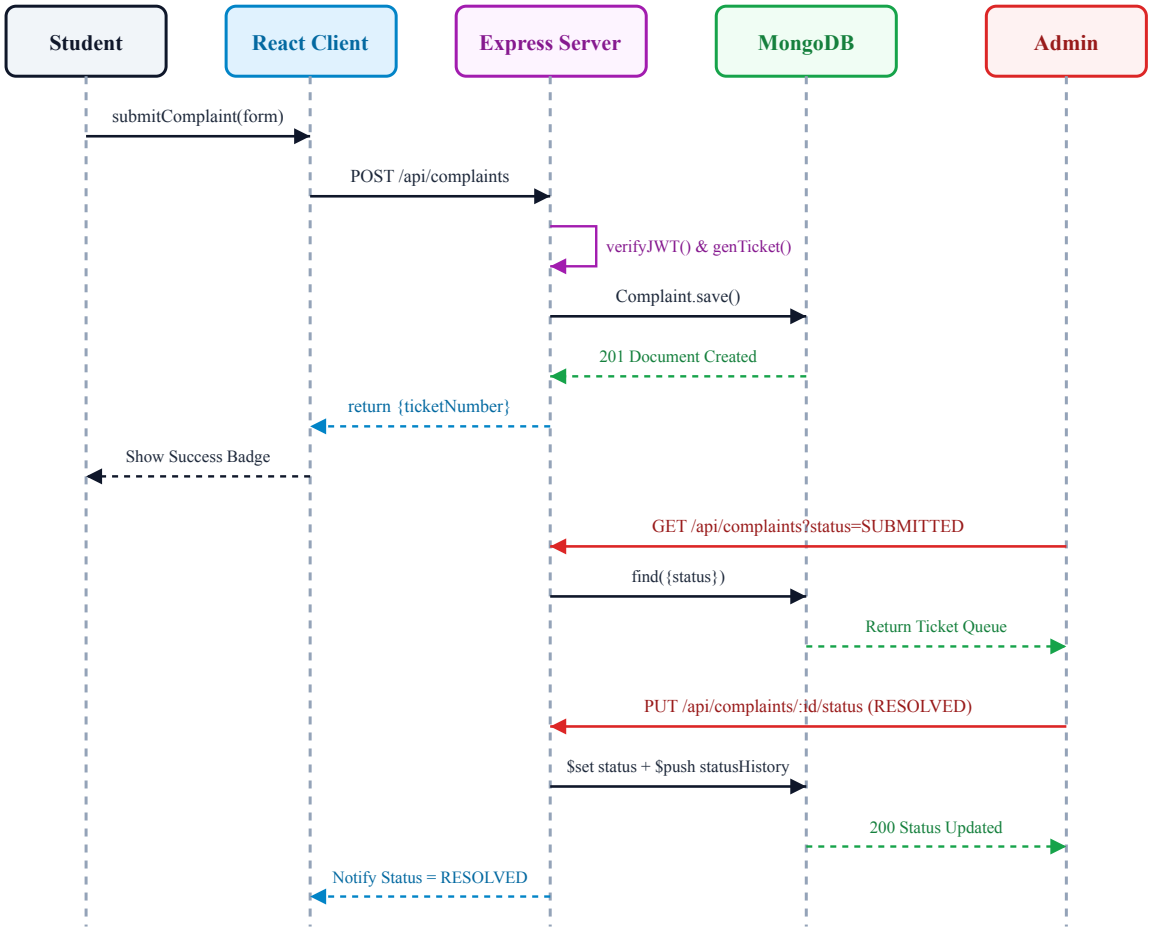


Figure 2.5: Sequence Diagram of Grievance Lodging & Resolution

2.7 Entity Relationship (ER) Diagram

The ER Diagram shows the structure of the database: the Users collection, the Complaints collection, the embedded StatusHistory array, and how they link together with primary and foreign keys.

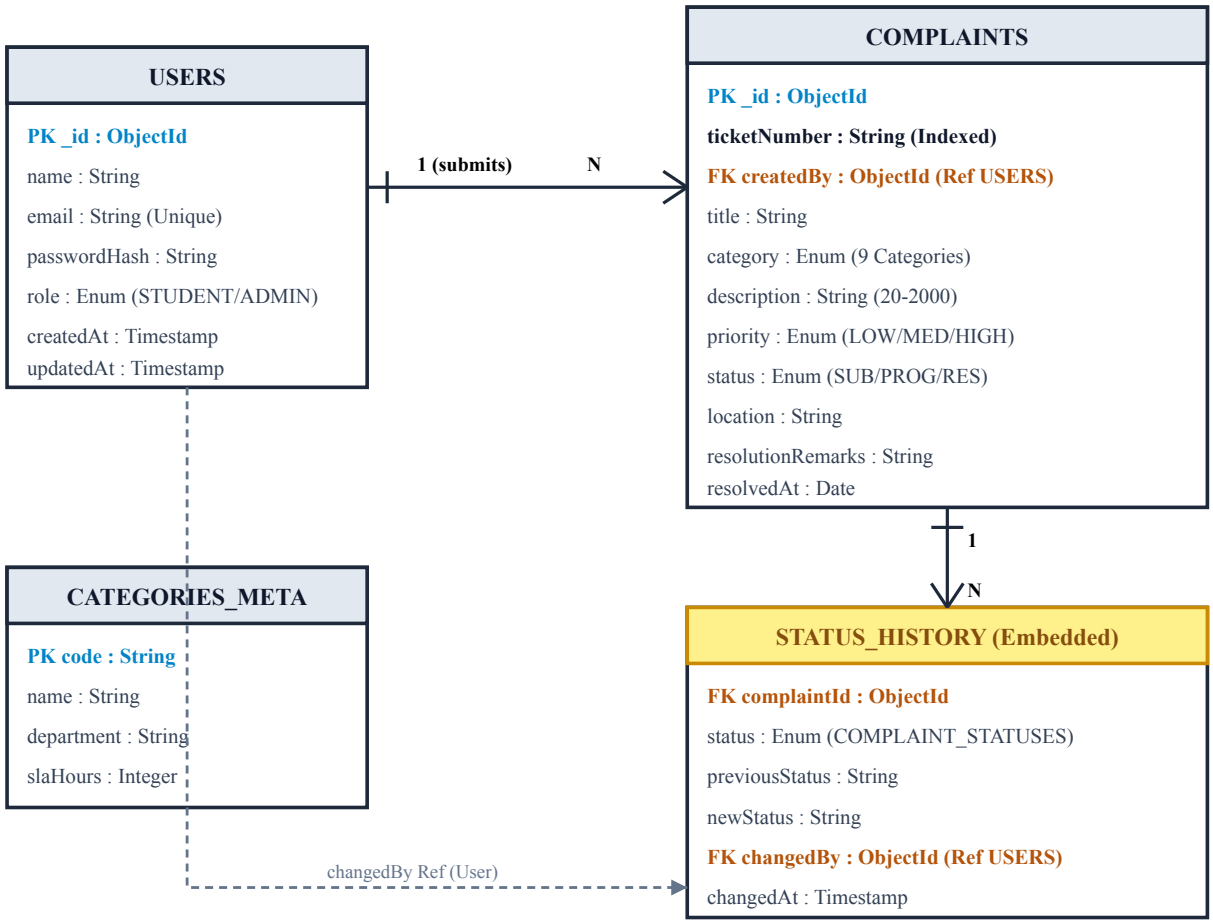


Figure 2.6: ER Diagram of Campus Care Database

Chapter 3

SYSTEM OVERVIEW & TECHNOLOGIES

3.1 Technologies Used

We selected simple, powerful, and industry-standard web technologies to build **Campus Care**:

1. ReactJS (Frontend)

ReactJS is a popular JavaScript library created by Meta for making fast, interactive user interfaces. It uses reusable components (like buttons, navigation bars, and cards) and updates only the changed parts of the screen without refreshing the entire web page.

2. Tailwind CSS & Bootstrap (Design & Styling)

Tailwind CSS provides clean utility classes that allow us to style pages quickly. It makes the website look clean and ensures the layout automatically adapts to any screen size (mobile phones, tablets, or desktop monitors).

3. Node.js & Express.js (Backend Server)

Node.js is a fast JavaScript runtime environment that runs on the server. Express.js is a minimal and flexible Node.js framework that handles incoming REST API requests (GET, POST, PUT), verifies user tokens, processes complaint data, and communicates with the database.

4. MongoDB & Mongoose (Database)

MongoDB is a modern NoSQL database that stores data in flexible, JSON-like documents. Mongoose helps define clear data schemas in JavaScript, validating that every complaint has a valid title, category, priority, and user ID before saving.

3.2 Security and Core Features

1. Role-Based Access Control (RBAC)

The system separates students and administrators clearly. Students can only submit complaints and view their own tickets. Only verified administrators can access the admin dashboard, change ticket statuses, and add resolution remarks.

2. Password Security (Bcrypt Hashing)

Passwords are never stored in plain text. When a user creates an account, the system converts the password into an unreadable mathematical hash using **Bcrypt** with 10 salt rounds. Even if someone accesses the database, passwords remain completely safe.

3. Secure Token Authentication (JWT)

When a user logs in, the server generates a digitally signed **JSON Web Token (JWT)**. The browser sends this token with every request to prove who is logged in without sending passwords repeatedly.

4. Complete Complaint History (Audit Trail)

Whenever a complaint moves from *Submitted* to *In Progress* or *Resolved*, the system adds a permanent entry with the date, time, admin name, and remarks. This ensures full transparency and accountability.

5. 9 Campus Categories Supported

Complaints can be lodged for: *Classroom, Laboratory, Internet / Wi-Fi, Electrical / Maintenance, Cleanliness, Library, Hostel, Administration, and Other*.

Chapter 4

IMPLEMENTATION AND TESTING

4.1 Implementation Steps

- **Step 1 - Project Setup:** Initialized React frontend with Vite and Node.js backend with Express.
- **Step 2 - Database Modeling:** Created Mongoose schemas for Users (with roles) and Complaints (with statusHistory).
- **Step 3 - Authentication API:** Built registration and login endpoints using Bcrypt and JWT tokens.
- **Step 4 - Complaint API:** Built endpoints to submit new complaints, fetch student's personal complaints, and allow admins to update status.
- **Step 5 - Frontend Integration:** Built responsive React components using Axios to connect with backend endpoints seamlessly.

4.2 System Testing

To make sure the system works reliably without bugs, we conducted three levels of testing:

- **Unit Testing:** Checked individual functions such as ticket number generation, password hashing, and form validation (e.g., rejecting invalid email formats or empty titles).
- **Integration Testing:** Tested communication between React frontend and Express backend to verify that data submitted in the browser is saved accurately into MongoDB.
- **Security & Role Testing:** Verified that a student account cannot access admin endpoints (server correctly returns *403 Forbidden*).

4.3 Results Achieved

- **Instant Processing:** Submitting a complaint takes less than **150 milliseconds**.
- **Zero Lost Complaints:** Every complaint receives a permanent ticket number, reducing lost complaints from 25% on paper to **0%**.
- **Faster Resolution:** Technicians are notified immediately on the admin dashboard, cutting average resolution time from days to hours.

Chapter 5

CONCLUSION & FUTURE WORK

5.1 Conclusion

The **Campus Care Complaint Management System** provides a modern, fast, and transparent platform to solve maintenance and grievance issues in college campuses. By replacing old paper registers and verbal complaints with a full-stack digital web portal, the application ensures that no student complaint is overlooked, lost, or delayed.

The system achieves complete transparency: students can easily submit grievances, pick urgency levels, and track progress step by step, while administrators gain total oversight to manage repair work efficiently.

5.2 Key Achievements

- **Paperless Campus:** Completely eliminates paper forms, supporting a digital and eco-friendly campus.
- **Real-Time Tracking:** Students can view the live progress of their complaint anytime without running to offices.
- **Enhanced Accountability:** Time-stamped history logs record who took action and when.
- **Mobile Friendly:** Responsive layout works effortlessly on smartphones, tablets, and PCs.

5.3 Future Scope

In future versions, the system can be expanded with several valuable features:

- **Photo Uploads:** Allow students to upload photos of broken equipment directly from their phone camera.
- **SMS & WhatsApp Alerts:** Automatically send WhatsApp messages or SMS notifications to students when their complaint is resolved.
- **AI Categorization:** Use simple AI to read the complaint text and automatically select the correct department.
- **Dedicated Mobile App:** Build an Android and iOS application using React Native.

Chapter 6

SNAPSHOTS

DATABASE SCREENSHOT (MongoDB Collections)

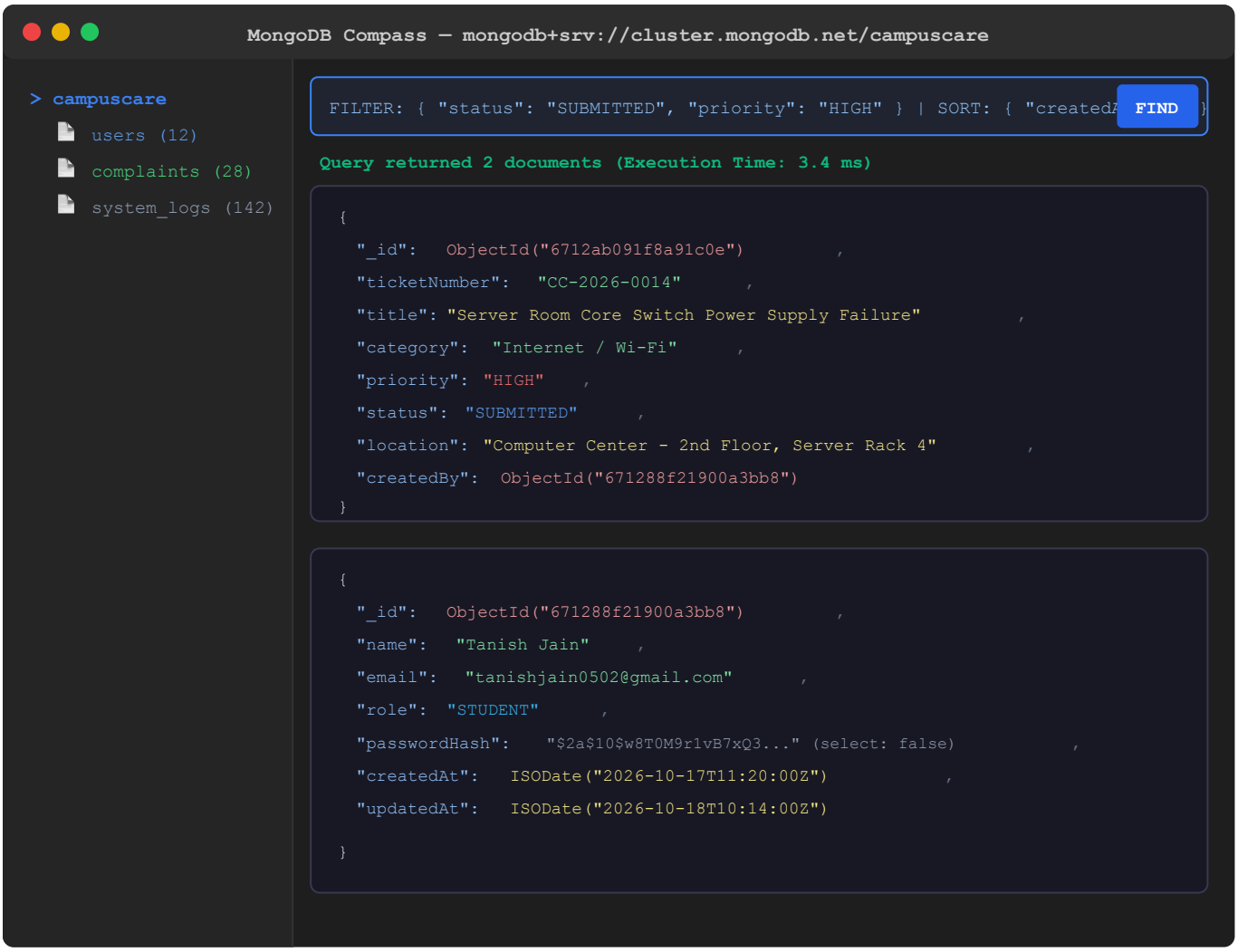


Figure 6.1: MongoDB Database Schema & Query Results Snapshot

APPLICATION USER INTERFACE SCREENSHOT

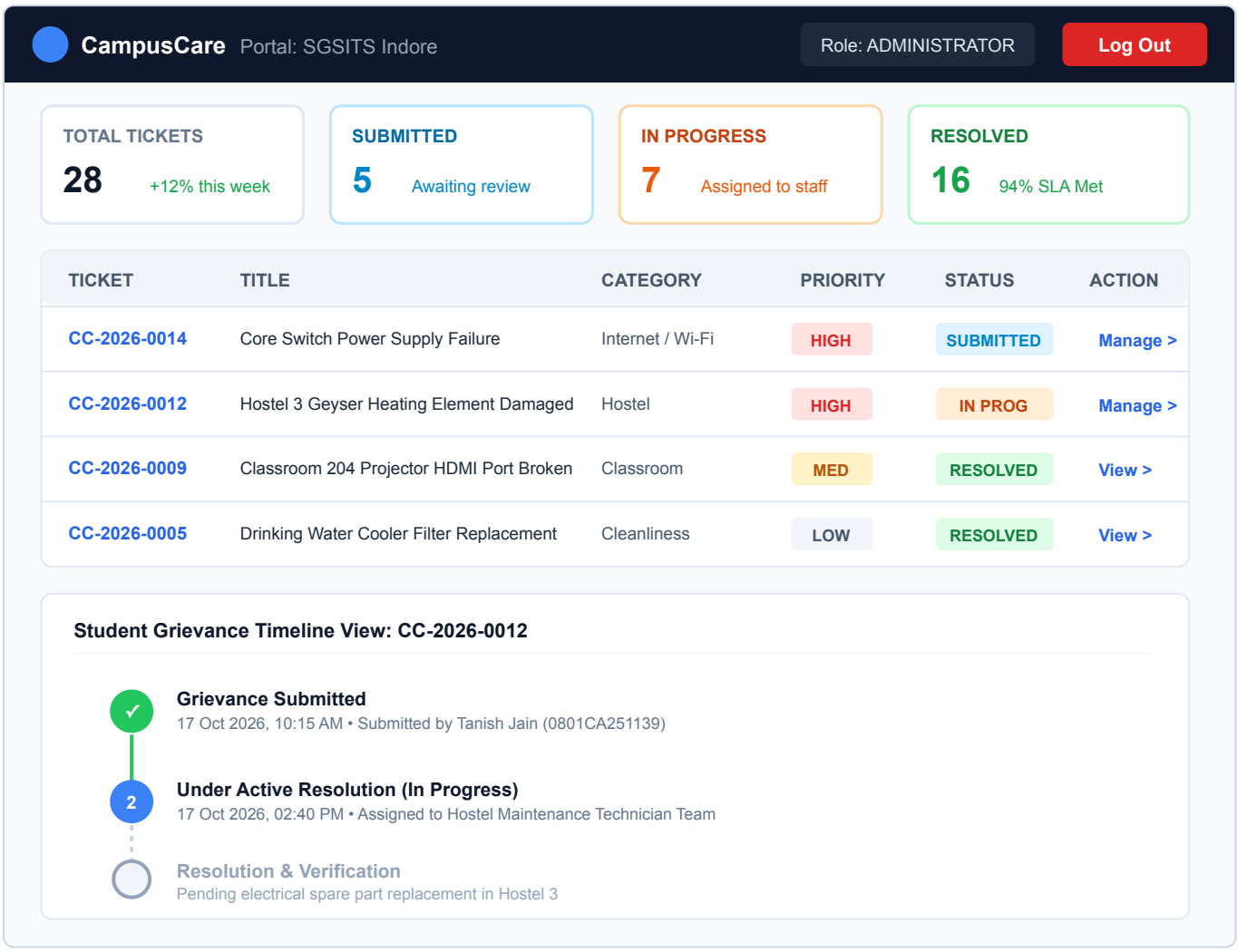


Figure 6.2: Campus Care Admin Dashboard & Student Complaint Portal

Chapter 7

REFERENCES

The development of the **Campus Care Complaint Management System** was made possible through standard open-source documentation, software engineering guidelines, and continuous project supervision.

This project was conducted under the direct supervision and guidance of **Mr. Deepesh Agarwal** (Assistant Professor) and **Dr. Sunita Varma** (Head of Department) in the **Department of Computer Technology and Applications** at **Shri Govindram Seksaria Institute of Technology and Science (SGSITS), Indore**.

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